

Midterm Review

STA 4724

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- 75 minutes in class midterm
- 20pt short answer questions
- 10pt (5pt*2) long questions
- A simple calculator is allowed
- One 3"x5" double-side ***handwritten*** cheat sheet is allowed
 - I will take away larger or printed cheat sheets

- Cover everything in lectures until Classification 3.
- Programming questions are limited to the functions taught in Python Basic 1,2,3

Sample Programming Question

- Q: Generate a random odd integer between 3 and 9.
- A: `random.randint(0,4) * 2 + 1`

Sample Concept Question

- Q: Given a vector $v = [1, 2, 3]$, compute $\|v\|_1$ and $\|v\|_2$.
- A: $\|v\|_1 = 1 + 2 + 3 = 6$
 $\|v\|_2 = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$
- For True/False Question, explain the reason if you answer False. You don't need to justify the answer if you answer True.

Sample Long Question

(5 points) Consider the confusion matrix

$$\begin{array}{c} \text{Predicted class} \\ \text{Actual class} \left[\begin{array}{cc} & \end{array} \right] \end{array}$$

compute Recall, Fallout, Specificity, Precision, and Accuracy.